Marie Curie

Polish-French physicist and chemist Marie Curie (1867–1934) overcame traditional barriers to women in science to make major contributions to physics, chemistry, and medicine. She discovered new chemical elements, developed scientific understanding of radioactivity, and founded two world-famous research labs—the Curie Institutes—in Paris and Warsaw.

Marie Curie's early life
The daughter of two schoolteachers, Marie (on the right here with her sister Hela) proved a bright student. She worked as a teacher and then a governess, before moving to France in 1891 to study physics and mathematics at the University of Paris. In 1906, she became the university's first female professor.

> "Be less curious about people and more curious about ideas."
>
> Marie Curie

Mother and daughter
Curie's eldest daughter, Irène (1897–1953) first worked with her mother as a radiographer during World War I. As a scientist in her own right, Irène Joliot-Curie along with her husband Frédéric won the 1935 Nobel Prize for the discovery of artificial radioactivity.

> "Marie Curie is, of all celebrated beings, the only one whom fame has not corrupted."
>
> Albert Einstein, 1934

Partnership with Pierre Curie

Born Maria Salomea Skłodowska in Warsaw, Poland, Marie moved to France to study at the University of Paris. There, she met French chemist Pierre Curie. The pair married in 1895 and began working together, studying the newly discovered phenomenon of radioactivity.

Discovering new elements

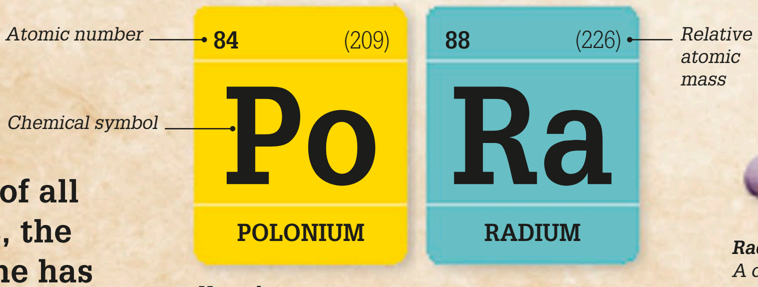
The Curies discovered that a uranium ore called pitchblende possessed higher levels of radioactivity than pure uranium and concluded it must contain other, more radioactive, substances. In 1898, after much painstaking work refining pitchblende, they discovered two previously unknown chemical elements—polonium and radium.

Winning the Nobel Prize

In 1903, the Curies, along with Henri Becquerel, won the Nobel Prize in Physics. Marie was the first female winner of the award. Pierre Curie died in a car accident in 1906, but despite this tragedy, Marie continued their work, managing to isolate pure radium in 1910. A year later, she received the Nobel Prize in Chemistry—the first person to win two Nobel prizes.

Wartime service

Marie Curie pioneered the use of radium to fight cancer tumors and helped develop the use of X-rays as a vital medical tool. When World War I began, she raised funds, organized, and even drove ambulances equipped with X-ray machines to battlefields. These were used to diagnose bullet and shrapnel injuries, saving thousands of lives.



New elements
Polonium and radium are the two chemical elements that the Curies discovered. Polonium is named after Marie's home country, Poland.

Radioactive flask
A clear, glass flask that Marie used in her work on radium has discolored and turned violet-blue after repeated exposure to radiation.